

ANN Practical 02 : XOR using Backpropagation Neural Network

Code

```
import numpy as np
from sklearn.neural_network import MLPClassifier

X = [[0,0],[0,1],[1,0],[1,1]]

y = [0,1,1,0]

mlp = MLPClassifier(
    hidden_layer_sizes=(2,),
    max_iter=1000,
    activation='logistic'
)

mlp.fit(X,y)

print("Predictions:", mlp.predict(X))
```

Aim

To implement the **XOR Logic Gate** using a **Backpropagation Neural Network**.

What is XOR?

XOR (Exclusive OR) gives:

Input A Input B Output

0	0	0
0	1	1
1	0	1
1	1	0

Rule:

- Inputs Different → Output = 1
- Inputs Same → Output = 0

Line-by-Line Explanation

Line 1

```
import numpy as np
```

Imports the NumPy library.

- Used for numerical calculations.
- In this code, NumPy is imported but not heavily used.

Viva: What is NumPy?

Answer: NumPy is a Python library used for numerical computations and arrays.

Line 2

```
from sklearn.neural_network import MLPClassifier
```

Imports MLPClassifier.

MLP = Multi-Layer Perceptron

It is a neural network model available in Scikit-Learn.

Viva: What is MLPClassifier?

Answer: MLPClassifier is a neural network classifier used for supervised learning problems.

Line 4

```
X = [[0,0],[0,1],[1,0],[1,1]]
```

Input dataset.

These are all possible combinations of XOR inputs.

A B

0 0

A B

0 1

1 0

1 1

Viva: What does X represent?

Answer: X represents the input features of the dataset.

Line 6

`y = [0,1,1,0]`

Expected XOR outputs.

A B Output

0 0 0

0 1 1

1 0 1

1 1 0

Viva: What does y represent?

Answer: y represents the target output or labels.

Line 8-12

```
mlp = MLPClassifier(  
    hidden_layer_sizes=(2,),  
    max_iter=1000,  
    activation='logistic'  
)
```

Creates the neural network.

hidden_layer_sizes=(2,)

Means:

- One hidden layer
- Contains 2 neurons

Architecture:

Input Layer (2 neurons)



Hidden Layer (2 neurons)



Output Layer (1 neuron)

Viva: Why is a hidden layer used?

Answer: Hidden layers help solve complex and non-linear problems.

max_iter=1000

Maximum training iterations.

The network can train up to 1000 times.

Viva: What is max_iter?

Answer: It specifies the maximum number of training iterations.

activation='logistic'

Uses the Sigmoid activation function.

Sigmoid Formula:

Output range:

- 0 to 1

Viva: Which activation function is used here?

Answer: Sigmoid (Logistic) activation function.

Line 14

```
mlp.fit(X,y)
```

Trains the neural network.

What happens internally?

1. Inputs are fed into the network.
2. Output is calculated.
3. Error is found.
4. Backpropagation updates weights.
5. Process repeats until error is minimized.

Viva: What does fit() do?

Answer: fit() trains the neural network using the given dataset.

Line 16

```
print("Predictions:", mlp.predict(X))
```

Predicts XOR outputs.

Expected Output:

Predictions: [0 1 1 0]

The predicted values match the XOR truth table.

Viva: What does predict() do?

Answer: It generates output predictions for new input data.

Program Output

Predictions: [0 1 1 0]

Meaning:

Input Predicted Output

[0,0] 0

[0,1] 1

[1,0] 1

Input	Predicted	Output
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[1,1]	0	
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Most Expected Viva Questions

Q1. What is Artificial Neural Network (ANN)?

Answer: ANN is a computing model inspired by the human brain that learns patterns from data.

Q2. What is Backpropagation?

Answer: Backpropagation is a learning algorithm that updates weights by propagating error backward through the network.

Q3. What is XOR?

Answer: XOR gives output 1 when inputs are different and 0 when inputs are the same.

Q4. Why is XOR a non-linear problem?

Answer: XOR cannot be separated using a single straight line.

Q5. Why can't a single-layer perceptron solve XOR?

Answer: Because XOR is non-linear and requires a hidden layer.

Q6. What is the role of hidden layers?

Answer: Hidden layers help learn complex patterns and relationships.

Q7. What is MLP?

Answer: MLP stands for Multi-Layer Perceptron.

Q8. What is an activation function?

Answer: An activation function decides the output of a neuron and introduces non-linearity.

Q9. Which activation function is used in this practical?

Answer: Sigmoid (Logistic).

Q10. What is the output range of Sigmoid?

Answer: Between 0 and 1.

Q11. What does fit() do?

Answer: It trains the neural network.

Q12. What does predict() do?

Answer: It predicts outputs using the trained model.

Q13. What is supervised learning?

Answer: Learning using input data along with known outputs (labels).

Q14. What library is used for ANN in this practical?

Answer: Scikit-Learn.

Q15. What is the final output of this program?

Answer: Predictions: [0 1 1 0]

One-Line Viva Summary

"This program uses an MLP (Multi-Layer Perceptron) neural network with backpropagation and sigmoid activation to learn and predict the XOR logic function."

Most Important Viva Questions

Q1. What is ANN?

Answer: ANN is an Artificial Neural Network inspired by the human brain.

Q2. What is Backpropagation?

Answer: Backpropagation is a supervised learning algorithm that updates weights by propagating errors backward.

Q3. What is XOR?

Answer: XOR gives output 1 when inputs are different and 0 when inputs are the same.

Q4. Why is XOR non-linear?

Answer: XOR data cannot be separated by a single straight line.

Q5. Can a single-layer perceptron solve XOR?

Answer: No, because XOR is a non-linear problem.

Q6. Why is a hidden layer used?

Answer: Hidden layers help solve complex and non-linear problems.

Q7. What is MLPClassifier?

Answer: MLPClassifier is a Multi-Layer Perceptron classifier available in Scikit-Learn.

Q8. Which activation function is used?

Answer: Sigmoid (Logistic) activation function.

Q9. What does fit() do?

Answer: It trains the neural network.

Q10. What does predict() do?

Answer: It predicts outputs using the trained model.

Q11. What is supervised learning?

Answer: Learning using input data and known output labels.

Q12. What is the output of XOR for inputs (1,1)?

Answer: 0

Q13. What is the output of XOR for inputs (0,1)?

Answer: 1

Q14. What libraries are used in this practical?

Answer: NumPy and Scikit-Learn.

Q15. What is the final result of this practical?

Answer: The neural network successfully learns and predicts the XOR function.

30-Second Exam Summary

"In this practical, we implemented the XOR function using a Multi-Layer Perceptron (MLP) neural network. XOR is a non-linear problem, so a hidden layer with 2 neurons and the Sigmoid activation function were used. The network was trained using Backpropagation and successfully predicted the outputs [0, 1, 1, 0]."

Experiment No. 2: XOR using Backpropagation Neural Network

Aim

To implement the XOR logical function using a Backpropagation Neural Network.

Theory (Short Viva Answer)

- ANN (Artificial Neural Network) is inspired by the human brain.
- Backpropagation is a supervised learning algorithm used to train neural networks.
- XOR is a non-linear problem.
- A single-layer perceptron cannot solve XOR.
- Therefore, a hidden layer is required.

XOR Truth Table

A B Output

0 0 0

0 1 1

1 0 1

1 1 0

Neural Network Architecture

Input Layer (2 Neurons)



Hidden Layer (2 Neurons)



Output Layer (1 Neuron)

Sigmoid Activation Function

- Converts input values into outputs between 0 and 1.

- Introduces non-linearity into the network.
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Code Explanation

Import Libraries

```
import numpy as np
```

Imports NumPy library.

```
from sklearn.neural_network import MLPClassifier
```

Imports MLPClassifier used to create the neural network.

Define Input Data

```
X = [[0,0],[0,1],[1,0],[1,1]]
```

All possible XOR input combinations.

Input A Input B

```
0      0
```

```
0      1
```

```
1      0
```

```
1      1
```

Define Output Data

```
y = [0,1,1,0]
```

Expected XOR outputs.

Create Neural Network

```
mlp = MLPClassifier(  
    hidden_layer_sizes=(2,),  
    max_iter=1000,
```

```
    activation='logistic'  
)
```

Parameters

hidden_layer_sizes=(2,)

One hidden layer containing 2 neurons.

max_iter=1000

Maximum 1000 training iterations.

activation='logistic'

Uses Sigmoid activation function.

Train Model

```
mlp.fit(X,y)
```

Trains the neural network using XOR data.

Backpropagation:

1. Input is given.
 2. Output is calculated.
 3. Error is calculated.
 4. Error is propagated backward.
 5. Weights are updated.
-

Predict Output

```
print("Predictions:", mlp.predict(X))
```

Predicts outputs for all XOR inputs.

Output

Predictions: [0 1 1 0]

Matches the XOR truth table.

